

Yeast Fermentation and Fuel Sources

Teacher guide | Biology 2e, Chapter 7 | 45-60 minutes

Learning targets and timing

How does the available sugar affect gas accumulation in a yeast culture?

Compare baker's yeast with no added sugar, glucose, and sucrose. Record cumulative gas volume and compare average accumulation rates over the same time interval.

Suggested sequence: 8-10 min prediction and setup; 20-30 min observation; 12-20 min analysis. Virtual time can be accelerated. For photosynthesis and yeast, pool independent replicates from class groups; three sequential repeats need extra class time.

Route choices and sharing

In Teacher setup, select Virtual or Classroom for each treatment/field and create an assignment link. Assign all virtual for absences or a substitute. For a hybrid route, conduct one part in class and simulate the others; compare data-source limitations before pooling. These links are editable and do not restrict access.

Materials

- Baker's yeast, glucose, sucrose, water, balance, thermometer and timer
- Three matched flasks and tubing connected to freely moving 100 mL gas syringes
- Teacher-prepared 30 deg C water bath; matched mixtures at 50 mL final volume

Classroom procedure

1. Label matched setups no added sugar, glucose, and sucrose. Prepare each sugar mixture with 2 g sugar; the water control receives no added sugar.
2. Add 0.5 g yeast to each and adjust to 50 mL final volume. Use the same yeast batch and equilibration time. Maintain 30 deg C with a water bath.
3. Connect each flask to a freely moving gas syringe. Check for leaks and unobstructed motion. Record the initial gas reading and subtract it from subsequent readings.
4. Record cumulative gas at 0, 5, 10, ... 30 minutes. Do not compare foam height or balloon circumference with the app's milliliter values.
5. Repeat with fresh mixtures for trials 2 and 3. Pool independent trials from class groups for a single-period lab; sequential repeats require additional time. Compare volume change divided by elapsed time, retain outliers, and explain suspected problems.

Safety: Use baker's yeast only. Do not taste cultures. Never seal gas production in a rigid closed container or block the syringe plunger. Stop before apparatus capacity is reached. Teacher checks glassware and warm-water handling.

Yeast Fermentation and Fuel Sources

Teacher guide / interpretation | Biology 2e, Chapter 7 | 45-60 minutes

Expected reasoning and troubleshooting

Virtual expectation: both sugar treatments accumulate more gas than the no-added-sugar control. Glucose and sucrose may exchange rank across trials. Encourage evidence-based comparison rather than a fixed "winning sugar."

Fermentation regenerates NAD⁺ so glycolysis can continue. Fermentation and anaerobic respiration are different pathways; yeast gas volume does not uniquely establish which metabolic processes occurred.

No-added-sugar is not necessarily zero because yeast has internal reserves and apparatus has background effects. A no-yeast control and leak/temperature checks strengthen the inference.

Model boundaries

Volumes are illustrative, generated curves with a lag and trial variation. Glucose and sucrose responses overlap; their ranking is not universal. The water control can show small background gas. The model does not establish anaerobic conditions or identify collected gas chemically.

Average gas accumulation rate = (final volume - initial volume) / elapsed minutes. Gas from yeast is consistent with CO₂ production, but volume alone does not prove fermentation rather than aerobic metabolism or directly measure ATP yield.

Suggested assessment - 12 points

Dimension	Evidence to look for	Points
Design	Reasoned prediction; variables, controls or sampling plan	0-3
Data	Complete observations, correct units, source labels, repeats	0-3
Analysis	Appropriate calculations and comparisons supported by data	0-3
Explanation	Biological reasoning, uncertainty, and a feasible improvement	0-3

For each dimension: 0 = absent; 1 = partial or major errors; 2 = mostly sound with a minor omission; 3 = complete and well supported. Suggested teacher rubric only; the app does not assign a grade.

Downloads, access, and classroom pilot

Student worksheet: three investigation pages, plus companion cutouts where included. Teacher guide: procedures, interpretation, and rubric. Reports and CSV data download from Analyze & explain. Teacher guides are public resources, not protected answer keys. Before classroom use, pilot the equipment, timing, and reading level. Editable Word and fillable PDF versions are planned.

Reading and method connections

OpenStax Biology 2e, 7.5: Open source

The investigation materials are original; appended cutouts were supplied by the classroom teacher. Simulated observations are not measurements from the linked sources.

Cut & sort: matter and energy

Optional introduction or extension | 10-15 minutes | Biology 2e, chapters 7 and 8.

Choose Card Activity on the lab overview to open the separate circular photosynthesis and respiration model, or arrange the printed cards on chart paper. Use two molecule sets for both halves at once, or reuse one 19-card set as you switch processes.

Print cutout pages in color, single-sided, at 100% scale. Cut along dashed lines. Use scissors safely. Group work: one person arranges, one counts atoms, and one explains; then rotate roles.

Model aerobic cellular respiration. Reuse the molecule cards from the photosynthesis sheets and add the three respiration symbols. Set aside any symbols not needed.

Count carbon, hydrogen, and oxygen atoms separately on each side. Explain how your arrangement shows conservation of matter and the transfer of energy. Energy symbols are annotations, not part of this simplified atom inventory.

This equation models aerobic respiration, not yeast fermentation. Fermentation does not use oxygen. Respiration begins in the cytoplasm; most later stages in plant and animal cells occur in mitochondria. Plants respire too.

Teacher use: assign this before prediction or after analysis. Accept either order of molecule groups on the same side. Ask students to explain placement, not just match a key. The online model checks each group and atom counts for practice feedback only; no grade is submitted.

The supplied originals follow, including answer keys. Student packets omit the photosynthesis teacher page and the answer text under the respiration cutouts. For respiration, the ATP symbol does not specify yield; ADP and phosphate are not shown.

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

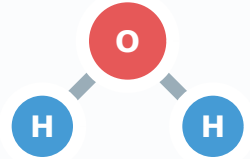
Build a balanced equation. Place light and chlorophyll where they help.



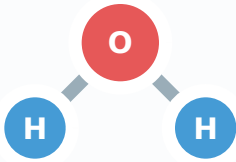
Oxygen



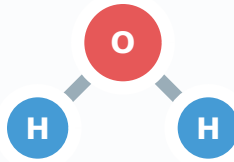
Carbon dioxide



Water



Water



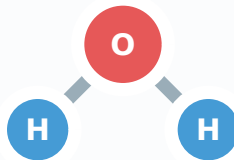
Water



Oxygen



Carbon dioxide



Water



Carbon dioxide

Atom colors: C = gray • O = red • H = blue

Student cutouts 1/3 • Print in color, single-sided, at 100%.

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

Build a balanced equation. Place light and chlorophyll where they help.

+



Combine with

O₂



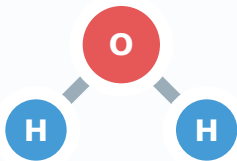
Oxygen

CO₂



Carbon dioxide

H₂O



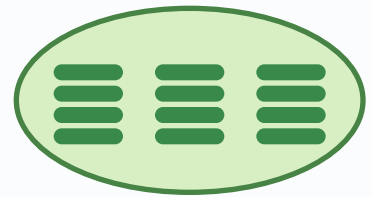
Water

+



Combine with

CHLOROPHYLL



Green pigment in chloroplasts
Simplified chloroplast diagram
Light-absorbing pigment

C₆H₁₂O₆



Atom inventory (not structure)

Glucose

O₂



Oxygen

LIGHT ENERGY



Sunlight

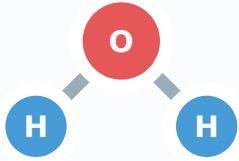
Atom colors: C = gray • O = red • H = blue

Student cutouts 2/3 • Print in color, single-sided, at 100%.

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

Build a balanced equation. Place light and chlorophyll where they help.



Water



Reaction arrow



Oxygen



Oxygen



Carbon dioxide



Carbon dioxide

Atom colors: C = gray • O = red • H = blue

Student cutouts 3/3 • Print in color, single-sided, at 100%.

TEACHER GUIDE

Photosynthesis: matter in, matter out, energy transferred

Copy: Print pages 1–3 for each group. Page 4 is for the teacher. Each complete set has 6 CO₂, 6 H₂O, 6 O₂, 1 glucose, 1 sun/light-energy card, 1 chlorophyll diagram, 2 plus signs, and 1 arrow.

Materials: Scissors, large chart paper, markers, and optional glue. Use sunlight or a lamp as a demonstration light source.

Student task: Sort the molecules into inputs and outputs. Use every molecule card to arrange a balanced photosynthesis equation. Place plus signs between molecule groups and the arrow between reactants and products. Place the sun/light-energy card and chlorophyll diagram above the arrow. Label the two sides “Reactants” and “Products.”

Answer key:



Light energy is absorbed by chlorophyll; show these above the arrow.

Check conservation: On each side, count 6 carbon atoms, 12 hydrogen atoms, and 18 oxygen atoms. The glucose picture is an atom inventory, not a structural model. Molecule pictures are simplified.

Short teacher script:

“Chlorophyll is a green pigment in chloroplasts that absorbs light energy.”

“Photosynthesis uses carbon dioxide and water to make glucose and oxygen.”

“Light provides energy. Chlorophyll helps absorb it; chlorophyll is not used up as a reactant.”

“Atoms are rearranged. Every atom on the reactant side must be accounted for on the product side.”

“Some of the absorbed light energy becomes chemical energy stored in glucose.”

Light-energy note: The single sun card represents the light energy needed for photosynthesis. Light energy is transferred and transformed; it is not an atom or a molecule. Do not include the sun or chlorophyll in the atom count.

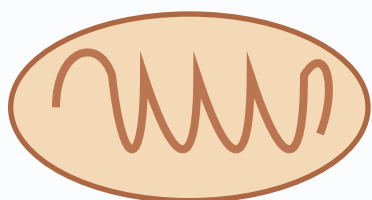
Quick discussion: Which cards represent matter? Which represent an energy source or an energy transfer? Why is chlorophyll placed above the arrow? How does the model show that matter is conserved?

CELLULAR RESPIRATION

Add-on cutouts • Use with your photosynthesis molecule cards

Cut on dashed lines. Each picture is a symbol, not a measured quantity.

MITOCHONDRION



Most stages occur here

Simplified organelle diagram

ATP



Energy transferred to ATP

Usable energy for cell work

Battery = symbol for energy storage

HEAT



Energy released as heat

Transferred to the surroundings

Build the respiration model: Reuse all 19 molecule cards. Arrange glucose and 6 oxygen molecules on the reactants side; arrange 6 carbon dioxide and 6 water molecules on the products side.

Matter equation:



Add the symbols: Place the mitochondrion above the arrow. Place ATP and heat beside the products to show energy transfer. Set the sun and chlorophyll cards aside. The energy cards are annotations; do not include them in the atom count.

Teacher sentence: “Cellular respiration transfers some of the chemical energy in glucose to ATP, which cells use to do work, and releases some energy as heat.”

Accuracy note: Respiration begins in the cytoplasm; most later stages in plant and animal cells occur in mitochondria. ATP is made from ADP and phosphate, which this simplified model does not show. One ATP card does not mean that only one ATP molecule is produced.

Check: Both sides contain 6 carbon, 12 hydrogen, and 18 oxygen atoms. Plants carry out cellular respiration too.

TEACHER INSTRUCTIONS

Cellular respiration • Reuse the photosynthesis molecule cards

Build the respiration model: Reuse all 19 molecule cards. Arrange glucose and 6 oxygen molecules on the reactants side; arrange 6 carbon dioxide and 6 water molecules on the products side.

Matter equation:



Add the symbols: Place the mitochondrion above the arrow. Place ATP and heat beside the products to show energy transfer. Set the sun and chlorophyll cards aside. The energy cards are annotations; do not include them in the atom count.

Teacher sentence: “Cellular respiration transfers some of the chemical energy in glucose to ATP, which cells use to do work, and releases some energy as heat.”

Accuracy note: Respiration begins in the cytoplasm; most later stages in plant and animal cells occur in mitochondria. ATP is made from ADP and phosphate, which this simplified model does not show. One ATP card does not mean that only one ATP molecule is produced.

Check: Both sides contain 6 carbon, 12 hydrogen, and 18 oxygen atoms. Plants carry out cellular respiration too.